

Nathaniel Martes

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EDUCATION

B.S. in Computer Science, Wilkes University, GPA: 3.95/4.00, Honors

Expected Graduation – May 2027

EXPERIENCE

Software Engineer Intern, NVIDIA

May 2026 – Current

- Contributed to the open-source distributed object storage system OpenStack Swift, that manages 1+ exabytes of storage and supports autonomous vehicle AI workflows that create 160M+ requests per day.
- Developed a Python middleware feature replicating AWS S3 checksum support, enabling clients to specify, verify, and persist object checksums throughout the object lifecycle, while creating comprehensive Pytest unit and functional tests to validate expected behavior and AWS S3 API compatibility, enabling faster and more reliable data workflows.
- Benchmarked checksum algorithms on distributed storage systems using Python and Ansible, identifying a 42% theoretical throughput improvement for production workloads by switching data integrity algorithms.
- Automated provisioning and teardown of distributed storage benchmarking environments using Ansible, Python, and Linux, reducing deployment time and improving testing efficiency.
- Reviewed 15+ open-source patches, providing code design feedback to senior and non-senior engineers.
- Collaborated with cross-functional teams to relay software release changes and new features for users and deployers.

Software Developer Intern, VetsEZ

Sep 2025 – May 2026

- Built a microservice with a team of developers using Java, Spring Boot, OData and Postman, which created a REST API for legacy SQL schemas that supported crucial platform modernization efforts.
- Utilized Python and Postman to automate regression testing of REST API remote producer call (RPC) execution, which saved the team 10 hours of manual regression testing.
- Integrated test cases into a GitHub Actions CI/CD pipeline for a SwaggerUI OpenAPI documentation generator using Java, successfully increasing generator code coverage to 91%.
- Optimized service efficiency by building a dedicated metadata caching layer using Java, Spring Boot, Docker, AWS ECS, and PostgreSQL, reducing startup latency for dependent APIs by 3 minutes.
- Cooperated with several developers using Microsoft Teams and Github Issues, improving code quality and coherence.

System Administrator Intern, Wilkes University

Jun 2024 – Current

- Automated server administration tasks using Ansible and Linux commands, saving 1 hour during software updates.
- Received mentorship from senior system administrators during on-boarding to understand server design and tools.
- Configured Python toolchains on workstations using Ansible, making it easier for students to debug programs.
- Migrated LDAP database to LDAPS using Bash, Ansible, and Linux, ensuring heightened security for user credentials.

DevSecOps Engineer Intern, Software Engineering Institute at Carnegie Mellon University

May 2025 – Aug 2025

- Used Python, Git, Docker, and a REST API to create a vulnerability detector for installed software packages across all internally managed systems, cutting potential vulnerability detection time from 30 days down to 1 day.
- Corrected 4,400+ SQLite entries with Python, achieving 100% accuracy in quarterly software license usage reports.
- Built a Python and REST API tool to detect duplicate software license usage, saving 2 hours of manual data analysis.
- Integrated 80+ test cases for internal Atlassian suite applications into DevOps CI/CD pipelines using Python, Pytest, and REST APIs, preventing deployment of misconfigured Docker containers.
- Collaborated in an Agile environment using Jira to track project progress and participate in daily Stand-Up meetings, enhancing team communication and improving overall project understanding.

PROJECTS

AWS Affiliated Hackathon 1st Place, Interactive AI Reflection Therapy Web Application

Oct 2025 – Oct 2025

- Created a serverless full stack application using JavaScript, React, HTML, Tailwind CSS, Python, Git, and DynamoDB that creates AI interactive narrative chapters for therapy and user resonance which placed 1st among 200+ participants.
- Utilized AWS Cloud Development Kit (CDK) to create infrastructure-as-code (IoC) which deployed a REST API Gateway and Lambda functions that interacted with a NoSQL database and a Large Language Model (LLM).
- Collaborated with 2 developers to design, develop, and deploy the complete application within a 24-hour period.

J.P. Morgan Chase Affiliated Hackathon 1st Place, AI Image Disaster Analyzer

Mar 2025 – Mar 2025

- Worked in a team of 4 to develop a full stack AI image analysis application for determining room disaster readiness using Python, Javascript, TypeScript, React, HTML, and CSS which placed 1st among 470+ participants.
- Built a REST API using FastAPI and Google Gemini to perform server side prompting and AI parsing.
- Established Git and Docker workflows which lead to better team efficiency under limited development time.

LEADERSHIP

Vice President, Mathematics, Physics, and Computer Science Club at Wilkes University

Jan 2025 – Current

- Managed hackathons for attendees to hackathons in University of Delaware and Leigh University, achieving a 1st place hackathon win and university recognition.
- Created general announcements using Discord for club information, meetings, and university events for over 60+ members.

Martial Arts Instructor, Dual Dragon Karate

May 2022 – Current

- Led hundreds of classes teaching self-defense techniques and traditional forms to 15+ students.
- Handled different student age groups from 4 years old to 50+, ensuring students are satisfied with the class and that they are learning real world skills.

Treasurer, Society of Professional Hispanic Engineers (SHPE) Wilkes Chapter

Dec 2025 – May 2026

- Gave proposal for national convention funding to 15+ student government members, resulting in a full funding amount of \$3800+.
- Kept track of funds from events and fundraisers using Google Sheets, ensuring organization at every step.

Public Relations Officer, Society of Professional Hispanic Engineers (SHPE) Wilkes Chapter

Jan 2025 – Dec 2025

- Managed LinkedIn social media page for Wilkes Chapter, displaying student success, community involvement, general updates to over 850+ users.

SKILLS

Programming: Python, TypeScript, JavaScript, Golang, SQL

Tools: Pytest, Spring Boot, FastAPI, React, HTML, CSS, Git, Docker, Linux, CI/CD, AWS, PostgreSQL, Ansible, Kubernetes